

The Future of Electronics Depends on Science

Electronics is a science—take any device and it ultimately boils down to the travelling of electrons, change in resistance or magnetism, transfer of energy, behaviour of atomic and sub-atomic particles, optics and so on. Understandably, disruptive changes in the field of electronics also often originate from science labs

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While electronics R&D labs do undeniably innovate, path-breaking findings coming from the pure sciences—new materials, chemical or physical properties, methods learnt from biological systems, etc.—often open up game-changing alternatives with the potential to disrupt the smooth flow of technological progress.

Telecommunications expert and faculty at the Illinois Institute of Technology, Dr Suresh Borkar says, “The primary focus in electronics and communications is miniaturisation, performance, universal access and power savings. The applications encompass almost all of human endeavour including communications, transportation, health management and entertainment. For continued progress, cross-disciplinary advances are needed in many areas including materials sciences, physical sciences, chemical sciences, biological sciences and mathematics.”

In the wake of the New Year, we decided to celebrate the contributions of science to the fields of electronics, computing and telecommunications, by discussing how pure science research could transform technology in the future. Here are some recent examples from a wide, unending pool.

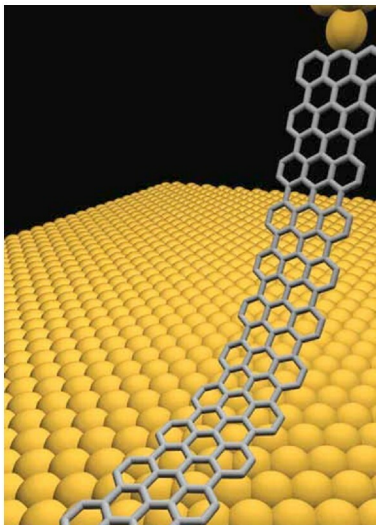
1 Intelligent self-learning systems

Research areas: Neuroscience, mathematics

Applications: Neural networks; self-learning systems for use in image recognition, speech systems, natural language modelling

Intelligence is always defined and benchmarked against man's sixth sense. Soft computing based on fuzzy logic and neural networks, uncertainty analysis and computational theory are all a result of extensive scientific research.

Neuroscientists across the world are trying hard to further understand the working of the human brain, quantify



Graphene nanoribbons (Image courtesy: naukoved.ru/news/view/280)

and design it, so it can be used to make smarter computing systems. Developments in this space will have great ramifications in image recognition, speech systems and natural language modelling.

There are several examples of futuristic research in this space, Google's self-taught neural network being the recent one.

In June this year, Google announced results of one of its abstract researches. Several years ago, a group of Google scientists started working on a simulation of the human brain. They connected over 16,000 processors to make one of the world's largest neural networks, with over a billion